Ferdaws Qaem

Computer Engineering Student | AI and Software Engineering

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Fourth-year Computer Engineering student with hands-on experience building local AI applications, cross-platform mobile products, and embedded IoT systems. Completed Microsoft's AI Innovators Internship Program and an Erasmus+ exchange at VSB-Technical University of Ostrava. Strong foundation in Python, C/C++, Dart, object-oriented design, data structures, and algorithms.

EXPERIENCE

AI Innovators Internship Program - Participant

Summer 2026 | Remote

Microsoft Turkey

- Built an end-to-end Retrieval-Augmented Generation (RAG) assistant with Python, local LLMs, and Microsoft Foundry Local for cloud-independent document question answering.
- Implemented document ingestion, indexing, contextual retrieval, prompt design, and local model orchestration in a privacy-first AI pipeline.
- Completed the program capstone and received the Microsoft AI Innovators Internship Program certificate of completion.

CERTIFICATIONS

AI Innovators Internship Program — 17 Aug 2026
Microsoft Turkey

Fundamentals of Agents - Unit 1 — 29 Aug 2026
Hugging Face Agents Course

EDUCATION

B.Sc. Computer Engineering

09/2023 - Expected 07/2027

Istanbul Kültür University | Istanbul, Türkiye | GPA: 3.04/4.00

Relevant coursework: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, Machine Learning, Embedded Systems and IoT, Mobile Programming, Project Management.

Erasmus+ Exchange

09/2025 - 01/2026

VSB-Technical University of Ostrava | Czech Republic

Faculty of Electrical Engineering and Computer Science. International academic experience in a cross-cultural engineering environment.

TECHNICAL SKILLS

Languages	Python, C, C++, Dart, GDScript
AI and Data	RAG, local LLMs, Microsoft Foundry Local, prompt design, machine learning fundamentals
Mobile	Flutter, Dart UI, cross-platform development, widget composition
Embedded and IoT	Sensor integration, real-time polling, alert systems, hardware-software interfacing
Engineering	OOP, data structures and algorithms, event-driven design, Git, GitHub, VS Code

SELECTED PROJECTS

Local RAG AI Assistant

Python | RAG | Foundry Local

- Architected a fully local RAG assistant that ingests custom knowledge sources, retrieves relevant context, and generates grounded answers without external cloud AI services.
- Connected Microsoft Foundry Local, Python, and local LLMs in a privacy-first application pipeline and delivered the project as the AI Innovators program capstone.

Campus Lost and Found

Flutter | Dart | Python | OOP

- Built a cross-platform campus app for reporting lost items and browsing found-item listings with search and filtering.
- Owned the backend data model and item lifecycle logic from lost to found to claimed, strengthening state management and API design skills.

HPC Temperature Alert System

Embedded C | IoT | Supabase

- Designed a real-time monitoring system for high-performance computing environments using dual thermal sensors and configurable alert thresholds.
- Added a ring-buffer cache and exponential backoff to maintain reliable behavior during connectivity failures and intermittent backend access.

Risk Management Interactive Game

Godot | GDScript | PMBOK

- Translated PMBOK risk management concepts into branching decisions and a scoring model that simulates real project trade-offs.
- Completed the full development cycle from requirements and game-loop design through implementation, testing, and a working demonstration.

Algorithms and Data Structures Practice Suite

C | C++ | Problem Solving

- Implemented more than 20 algorithm and data-structure exercises covering linked lists, trees, sorting, recursion, and complexity analysis.
- Strengthened low-level reasoning in memory management, pointer arithmetic, correctness, and runtime trade-offs.

LANGUAGES

Persian: Native | English: Professional | Turkish: Intermediate

More work and demonstrations: ferdaws-c.github.io/portfolio